

MATTMAR V, Sweden

WMO: 022160

Lat: 63.3054N

Lon: 13.8767E

Elev: 1181

StdP: 14.08

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-4.7	-14.3	2.6	-10.9	-8.2	3.7	-4.3	24.7	35.2	22.1	32.2	0.7	320	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	76.2	61.9	72.7	60.2	69.0	58.5	63.5	72.8	61.6	70.1	59.8	67.1	7.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.8	80.2	67.1	58.0	75.0	64.9	56.3	70.6	63.3	29.4	72.4	28.1	70.4	26.7	67.2	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.8	16.4	DB	-16.1	80.9	6.4	3.5	-20.7	83.4	-24.5	85.4	-28.1	87.4	-32.7	89.9	(4)
(5)			WB	-16.1	65.8	6.1	2.6	-20.5	67.7	-24.0	69.2	-27.4	70.7	-31.8	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	19.5	21.2	27.7	37.0	45.0	52.9	58.1	55.2	47.3	37.6	29.2	22.3
	DBStd	15.63	11.81	11.66	8.48	6.04	6.48	6.57	5.65	5.26	5.52	6.94	8.31	11.92
	HDD50	5057	945	806	693	391	186	40	3	12	112	385	625	859
	HDD65	9927	1410	1226	1158	840	621	365	224	306	530	848	1075	1324
	CDD50	624	0	0	0	1	31	128	255	174	33	2	0	0
	CDD65	18	0	0	0	0	0	3	12	3	0	0	0	0
	CDH74	193	0	0	0	0	6	56	107	24	0	0	0	0
	CDH80	12	0	0	0	0	0	1	10	1	0	0	0	0
Wind	WSAvg	6.6	5.9	6.3	7.5	7.2	7.6	7.5	6.6	5.7	6.6	6.7	5.8	5.6
Precipitation	PrecAvg	25.5	1.7	1.4	1.4	1.4	1.7	2.8	3.7	3.3	2.2	2.2	2.0	1.7
	PrecMax	35.9	2.9	3.0	2.4	2.7	4.3	5.0	7.8	5.9	4.5	6.7	4.1	3.8
	PrecMin	18.9	0.2	0.2	0.3	0.3	0.5	0.7	0.8	1.4	0.7	0.6	0.4	0.7
	PrecStd	4.5	0.7	0.6	0.5	0.7	0.9	1.2	2.0	1.5	1.1	1.3	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	42.5	47.6	61.9	73.8	78.7	81.5	77.2	66.4	56.3	51.1	43.8
		MCWB	38.3	37.7	39.7	50.1	60.1	62.3	65.3	62.1	56.3	48.9	45.9	40.1
	2%	DB	39.7	40.0	43.7	55.5	67.6	75.5	77.0	72.2	62.1	52.5	46.3	41.4
		MCWB	36.3	36.5	38.4	45.6	55.0	61.2	62.8	61.0	54.0	46.8	42.6	37.9
	5%	DB	36.9	37.9	41.2	51.3	61.8	70.4	73.5	68.7	59.2	50.3	42.2	39.3
		MCWB	34.4	34.9	37.0	43.1	51.8	58.5	60.8	59.3	53.0	45.8	39.2	36.4
	10%	DB	34.5	35.5	38.9	47.5	56.7	65.6	69.8	65.3	56.4	48.0	39.4	36.8
		MCWB	32.3	32.9	35.4	40.9	48.7	56.1	59.1	58.1	50.9	44.8	36.9	34.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	38.6	41.5	50.8	60.6	63.9	66.4	64.8	58.2	51.3	46.5	40.6
		MCDB	41.9	41.3	46.1	61.0	74.2	75.3	78.9	72.6	63.4	53.7	50.2	43.3
	2%	WB	36.7	36.9	39.3	46.7	55.9	61.6	64.3	62.5	55.5	48.1	42.9	38.3
		MCDB	39.2	39.6	42.9	54.5	65.2	73.2	74.7	69.0	60.2	51.5	46.1	40.9
	5%	WB	34.7	35.1	37.3	44.0	52.8	59.4	62.3	60.5	53.4	46.5	39.7	36.5
		MCDB	36.9	37.5	40.8	49.9	60.3	69.3	71.0	66.6	58.0	49.5	41.9	38.9
	10%	WB	32.2	32.9	35.6	41.7	49.6	56.7	60.3	58.8	51.6	44.6	37.1	34.4
		MCDB	34.1	35.2	38.7	46.6	56.4	64.6	67.6	64.4	55.6	48.0	39.0	36.5
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.6	13.3	14.6	15.4	15.6	15.5	14.6	12.6	10.3	8.8	10.0
		MCDBR	10.9	11.0	13.4	21.8	23.9	23.8	21.8	20.4	17.4	13.7	12.5	11.1
		MCWBR	9.1	9.1	9.3	13.2	13.5	11.9	10.0	10.8	10.5	9.5	9.8	9.1
	5% WB	MCDBR	11.3	10.8	12.2	19.9	22.3	22.3	20.5	17.5	15.2	12.6	12.4	10.4
		MCWBR	9.8	9.3	9.0	12.5	13.1	11.6	10.3	10.2	9.8	9.5	10.1	8.8
Clear-Sky Solar Irradiance	taub	0.200	0.262	0.292	0.324	0.332	0.329	0.342	0.331	0.303	0.286	0.211	0.185	
	taud	2.035	2.245	2.307	2.397	2.477	2.515	2.491	2.512	2.574	2.480	2.112	1.988	
	Ebn at Noon	141	219	258	273	281	284	276	270	258	213	134	74	
	Edh at Noon	9	18	26	30	30	29	29	26	20	15	8	4	
All-Sky Solar Radiation	RadAvg	56	233	666	1228	1470	1606	1521	1147	685	295	82	27	
	RadStd	6	32	70	100	121	133	171	105	67	38	11	2	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page